

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

AGROFRESH SOLUTIONS, INC.,
Petitioner,

v.

LYTONE ENTERPRISE, INC.,
Patent Owner.

IPR2021-00451
Patent 6,897,185 B1

Before SUSAN L. C. MITCHELL, ZHENYU YANG, and DEVON
ZASTROW NEWMAN, *Administrative Patent Judges*.

NEWMAN, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

INTRODUCTION

A. Background and Summary

AgroFresh Solutions, Inc., (“Petitioner”) filed a Petition pursuant to 35 U.S.C. §§ 311–319 requesting an *inter partes* review of claims 1–9 and 11–15 (“the Challenged Claims”) of U.S. Patent No. 6,897,185 B1 (Ex. 1001, “the ’185 patent”). Paper 1 (“Pet.”). Lytone Enterprise, Inc., (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 6 (“Prelim. Resp.”). With prior authorization, Petitioner filed Petitioner’s Reply to Patent Owner’s Preliminary Response (Paper 7, “Reply to Prelim. Resp.”), and Patent Owner filed Patent Owner’s Preliminary Sur-reply (Paper 8, “Prelim. Sur-reply”).

Because Patent Owner had filed a Statutory Disclaimer of claims 1, 2, 4–10, and 12–15 shortly before the Petition was filed (Exhibit 2004), we considered at institution whether to institute review on the remaining Challenged Claims 3 and 11 as set forth in the remaining Grounds 2 and 3. Pet. 8.

On July 29, 2021, we issued a Decision ordering that “an *inter partes* review of claims 3 and 11 of U.S. Patent No. 6,897,185 B1 is instituted with respect to Grounds 2 and 3 of the Petition that address the patentability of claims 3 and 11.” Paper 11, 37 (“Dec.”). After institution, Patent Owner filed a Patent Owner’s Response (Paper 19, “PO Resp.”). Petitioner filed a Petitioner’s Reply to Patent Owner’s Response (Paper 21, “Reply”). Patent Owner filed Patent Owner’s Sur-Reply (Paper 22, “PO Sur-reply”). Patent Owner and Petitioner presented oral arguments on May 2, 2022, and a transcript has been entered into the record. Paper 28 (“Tr.”).

The Board has jurisdiction under 35 U.S.C. § 6. In this Final Written Decision, after reviewing all relevant evidence and assertions, we determine

that Petitioner has met its burden of showing, by a preponderance of the evidence, that claims 3 and 11 of the '185 patent are unpatentable.

B. Real Parties-in-Interest

Petitioner identifies as real parties-in-interest AgroFresh Solutions, Inc., AF Solutions Holdings LLC, and AgroFresh Inc. Pet. 5.

Patent Owner identifies Lytone Enterprise, Inc., as the real party-in-interest. Paper 4, 1.

C. Related Matters

Petitioner and Patent Owner both identify that the '185 patent is a subject of *Lytone Enterprise, Inc. v. AgroFresh Solutions, Inc.*, Case No. 1:20-cv-00678-LPS, D. Del. Pet 5; Paper 4, 1.

D. The '185 Patent (Ex. 1001)

The '185 patent, titled "Formulation for Counteracting and Ethylene Response in Plants, Preparation Process Thereof, and Method Using the Same," issued May 24, 2005, from Application No. 10/182,403, filed September 22, 2000. Ex. 1001, codes (21), (22), (45), (54). The formulation described by the '185 patent is "a novel tablet dosage formulation for regulating plant physiology" by counteracting the response of the plant to the growth hormone ethylene, which "affects many important aspects of plant growth, development, and senescence," the most important for commercial purposes being "the ripening of fruits, the senescence of flowers, and the abscission of leaves." *Id.* at 1:9–23. The formulation contains "an agent for blocking the ethylene binding site in plants and an effervescent ingredient, in admixture with one or more acceptable carriers and/or excipients." *Id.* at 2:19–21. The agent for blocking the ethylene binding site "includes all the conventional compounds that inhibit ethylene responses in plants, such as . . . 1-methylcyclopropene." *Id.* at 2:35–38. The

'185 patent discloses that gaseous 1-methylcyclopropene ("1-MCP") is effective at low dosage and safe for use on fruits, vegetables, and flowers.

However, 1-MCP is "highly unstable" and readily undergoes oxidation and other reactions. *Id.* at 1:51–52. The '185 patent further discloses that molecularly encapsulated forms of 1-MCP, such as those stabilized by a carrier, "provid[e] a convenient and safe means of storing, transporting and applying or delivering the gas to plants." *See id.* at 1:55–56. According to the '185 patent, these "[c]ommercial methylcyclopropene powder products usually contain 0.43% of a methylcyclopropene gas ingredient. The gas will be released after mixing the powder with water or a buffer solution." *Id.* at 1:59–60. When released in a sealed space at a final concentration of 900 parts per billion, the 1-MCP gas is effective "for at least 4 hours as a post-harvest treatment." *Id.* at 1:60–65.

E. Challenged Claims

Petitioner challenges claims 3 and 11 of the '185 patent. Claim 3 depends from claim 2, which depends from independent claim 1, both of which are not at issue in this proceeding. Ex. 1001, 6:11–23. Claim 11 also depends from claim 1. *Id.* at 6:45–48. Claims 1, 2, 3, and 11 are reproduced below:

1. An effervescent tablet dosage comprising an agent for blocking the ethylene binding site in plants and an effervescent ingredient, in admixture with one or more acceptable carriers and/or excipients.
2. The tablet dosage of claim 1, wherein the agent for blocking the ethylene binding site in plants is selected from the group consisting of cyclopropene, 1-methylcyclopropene, 3,3-dimethylcyclopropene, methylenecyclopropane, diazocyclopentadiene, transcyclooctene, cis-cyclooctene, and 2,5-norbornadiene, the derivatives thereof, and the mixtures thereof.

3. The tablet dosage of claim 2, wherein the agent for blocking the ethylene binding site in plants is 1-methylcyclopropene.

11. The tablet dosage of any one of claim 1, wherein the agent for blocking the ethylene binding site in plants is released in a gaseous form.

Ex. 1001, 6:11–49.

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 3 and 11 of the '185 patent would have been unpatentable based on the following grounds (*see* Pet. 8):

Ground ¹	Claims Challenged	35 U.S.C. §	References/Basis
2	3, 11	103(a) ²	Hisano, ³ Daly ⁴
3	3, 11	103(a)	Daly, Hisano

In support of its arguments, Petitioner relies on the Declaration of Krista S. Walton, Ph.D., (Ex. 1006) and the Supplemental Declaration of Krista S. Walton, Ph.D. (Exh. 1035).

In support of its arguments, Patent Owner relies on the Declaration of Randolph Beaudry, Ph.D. (Ex. 2013).

¹ Grounds 1 and 4 are obviated by Patent Owner's statutory disclaimer that cancelled all claims except 3 and 11.

² The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) ("AIA"), amended 35 U.S.C. § 103. Because the '185 patent has an effective filing date prior to the effective date of the applicable AIA amendments, we refer to the pre-AIA versions of § 103.

³ Hisano *et al.*, translation of Japanese Patent Application Publication No. H6-183903 ("Hisano"). The original Hisano reference is provided as Ex. 1003. (Pet. 33).

⁴ Daly *et al.*, U.S. 6,017,849 ("Daly")

G. Weight to Give Expert Testimony

Patent Owner argues that Dr. Walton lacks “firsthand experience” because she “has no experience in postharvest biology and admits she has not used compositions containing 1-MCP or conducted research or experimentation observing its release from molecular encapsulation complexes, whether at the time of the invention or at any time since.”

PO Resp. 16. Patent Owner submits testimony from Dr. Walton in a separate matter in which she indicated she had not been familiar with the properties of 1-MCP prior to that case. *Id.* (citing Ex. 2021 14:17–21). Patent Owner argues that Dr. Walton’s lack of direct experience with 1-MCP and experience in food storage should cause the Board to give her testimony little weight. *Id.* at 17.

Petitioner disagrees with Patent Owner’s characterization of Dr. Walton’s knowledge and notes that the unrelated litigation from which Patent Owner drew the testimony “did not relate to the use of a well-known α -cyclodextrin encapsulation complex to hold 1-MCP gas molecules, as disclosed in Daly and commercialized as EthylBloc prior to the alleged invention of the ‘185 Patent” or relate to creation of effervescent tablets using a 1-MCP/ α -cyclodextrin powder component. Reply 22–23. Petitioner further argues that Dr. Walton’s lack of research experience in the areas cited by Patent Owner are irrelevant because the starting point of the skilled artisan in this matter is a “known 1-MCP powder complex, which has already been stabilized” as shown by Daly and EthylBloc. *Id.* at 24.

Petitioner argued that Dr. Walton “has direct knowledge regarding the binding of gases and the formation of tablets, based on both her education and research in the field” and that neither Dr. Walton nor Patent Owner’s expert have provided any testimony to explain why 1-MCP encapsulated in

an α -cyclodextrin cage would be different from “any other gaseous guest molecule” or how a background in food storage would be necessary to opine on the technical issues in this matter. *Id.* at 24–25.

Patent Owner does not respond to these arguments, but asserts that Dr. Beaudry’s direct experience with 1-MCP uniquely qualifies him to testify on the technical issues in this case while Dr. Walton’s evidence is not credible and mere speculation. PO Sur-reply 1.

It is not necessary for an expert’s experience or expertise to precisely match the art at issue. *SEB S.A. v. Montgomery Ward & Co.*, 594 F.3d 1360, 1373 (Fed. Cir. 2010). However, one offering expert testimony must at least have ordinary skill in the art to provide relevant and reliable testimony that is helpful to the factfinder. *Kyocera Senco Indus. Tools, Inc. v. ITC*, 22 F.4th 1369, 1376–77 (Fed. Cir. 2022). “The Board has broad discretion to assign weight to be accorded expert testimony.” Consolidated Trial Practice Guide 35 (available at <https://www.uspto.gov/TrialPracticeGuideConsolidated>).

Dr. Walton has a Ph.D. in Chemical Engineering and nearly 20 years of experience in gas adsorption and separation and purification process design, with expertise in formation of shaped bodies like tablets from powdered materials, including in the commercial setting. Ex. 1006 ¶¶ 6–10; Ex. 1007. She is author or co-author on over 100 peer-reviewed journal articles involving chemical engineering. Ex. 1007, 2–9. We find this level of skill meets as well as exceeds the qualifications for the level of skill in the art under the training and experience portion of the definition we have adopted for purposes of this opinion (*see infra* § ANALYSIS, Section B.): A bachelor’s degree in chemistry, chemical engineering, materials science or the like, and 1–2 years of work experience, or a master’s degree in exchange

for such work experience, in the area of controlled release of compounds, such as those used to inhibit ethylene response in plants.

Patent Owner's expert Dr. Beaudry has a Ph.D. in horticulture and 35 years of experience studying controlled release of compounds used to optimize plant storage. Ex. 2014. He has extensively spoken and published on food storage and transport topics. *Id.* at 3–7; Ex. 2013 ¶¶ 5–6. We also find this level of skill meets as well as exceeds the qualifications for the level of skill in the art we have adopted herein. See *infra* ANALYSIS, Section B.

In reviewing Dr. Walton's testimony, as well as the testimony provided by Dr. Beaudry, we consider their opinions on the issues, compare their opinions to the disclosures of the prior art references and the challenged patent, and the entirety of the evidence of record to weigh each part of their opinion testimony separately. See *Elbit Sys. of Am., LLC v. Thales Visionix, Inc.*, 881 F.3d 1354, 1358 (Fed. Cir. 2018) (“The [Patent Trial and Appeal Board (‘PTAB’)] [i]s entitled to weigh the credibility of the witnesses.” (citation omitted)); *Icon Health & Fitness, Inc. v. Strava, Inc.*, 849 F.3d 1034, 1041 (Fed. Cir. 2017) (“To the extent [a party] challenges the PTAB's factual findings, ... the PTAB is permitted to weigh expert testimony and other record evidence and, in so doing, rely on certain portions of an expert's declaration while disregarding others.”).

ANALYSIS

A. Legal Standards

A claim is unpatentable for obviousness under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are “such that the subject matter as a whole would have been obvious at the time

the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) when in evidence, objective indicia of nonobviousness, i.e., secondary considerations⁵. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

Additionally, the obviousness inquiry typically requires an analysis of “whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue.” *KSR*, 550 U.S. at 418 (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006) (requiring “articulated reasoning with some rational underpinning to support the legal conclusion of obviousness”)); see *In re Warsaw Orthopedic, Inc.*, 832 F.3d 1327, 1333 (Fed. Cir. 2016) (citing *DyStar Textilfarben GmbH & Co. Deutschland KG v. C.H. Patrick Co.*, 464 F.3d 1356, 1360 (Fed. Cir. 2006)).

Although the prior art that can be considered in *inter partes* reviews is limited to patents and printed publications [under § 311], it does not follow that we ignore the skilled artisan’s knowledge when determining whether it would have been obvious to modify the prior art. Indeed, under 35 U.S.C. § 103(a), the obviousness inquiry turns not only on the prior art, but whether “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious . . . to a person having ordinary skill in the art to which said subject matter

⁵ Neither party introduces evidence of secondary considerations in this case. See generally Pet.; PO Resp.

pertains.” 35 U.S.C. § 103. Regardless of the tribunal, the inquiry into whether any “differences” between the invention and the prior art would have rendered the invention obvious to a skilled artisan necessarily depends on such artisan’s knowledge. *See Koninklijke Philips v. Google*, 948 F.3d 1330, 1337–39 (Fed. Cir. 2020).

B. Level of Ordinary Skill in the Art

We consider the asserted grounds of unpatentability in view of the understanding of a person of ordinary skill in the art. *KSR*, 550 U.S. at 398–399 (stating that obviousness is determined against the backdrop of the scope and content of the prior art, the differences between the prior art and the claims at issue, and the level of ordinary skill in the art). Factual indicators of the level of ordinary skill in the art include “the various prior art approaches employed, the types of problems encountered in the art, the rapidity with which innovations are made, the sophistication of the technology involved, and the educational background of those actively working in the field.” *Jacobson Bros., Inc. v. U.S.*, 512 F.2d 1065, 1071 (Ct. Cl. 1975); *see also Orthopedic Equip. Co. v. U.S.*, 702 F.2d 1005, 1011 (Fed. Cir. 1983) (quoting with approval *Jacobson Bros.*).

Petitioner contends that a person of ordinary skill in the art at the time would have had a bachelor’s degree in chemistry, chemical engineering, materials science or the like, and 1–2 years of work experience, or a master’s degree in exchange for such work experience, in the area of controlled release of compounds, such as those used to inhibit ethylene response in plants.

Pet. 19 (citing Ex. 1006 ¶¶ 13–20). Petitioner’s expert explains that the proposed level of education and experience was typical of scientists in the time frame, and was sufficient to understand and apply the technology

disclosed in the '185 patent, which is “not particularly sophisticated such that a person of ordinary skill in the art at the time of the invention would need additional education or work experience in order to understand it.”

Ex. 1006 ¶¶ 18–19.

Patent Owner disagrees that the skilled artisan would have only needed to have experience in the “general area of controlled release of compounds” and proposes that the person would have had

at least a bachelor’s degree in chemistry, chemical engineering, plant physiology, horticulture, plant science or the like, and at least 2-4 years of work experience, or a master’s degree in exchange for such work experience, in the area of postharvest biology and technology, including experience working with encapsulated gaseous agents such as the 1-MCP encapsulation complex, at the time of the invention in late 1999 and early 2000.

PO Resp. 14 (citing Ex. 2013 ¶¶ 15–16).

The central difference between these proposals is that Petitioner’s definition focuses on chemistry and materials science without a specific focus to the application of that knowledge to the subject matter of the invention, while Patent Owner’s proposal permits the artisan’s degree to be in horticulture or plant science, requires at least 2–4 years of experience in the area of postharvest biology and technology, and specifically requires “working with encapsulated gaseous agents such as the 1-MCP encapsulation complex.” PO Resp. 14. It is not clear on the record before us nor does Patent Owner’s expert explain why the additional, specific expertise in postharvest biology and technology would have been required to understand the teachings of the '185 patent in a manner necessary to apply them. Moreover, the proposed additional skill level does not appear to be necessary in light of the skill level reflected by the asserted prior art. *See*

Okajima v. Bourdeau, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not shown” (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163–64 (Fed. Cir. 1985)). For instance, Hisano measures preservation visually, by ascertaining whether the flowers looked fresh or appeared to have been harmed by effects of the chemicals added, and whether the water the flowers were stored in was cloudy. Ex. 1004 ¶ 24. This observation does not require specific expertise in postharvest biology and technology.

Petitioner’s assessment appears consistent with the level of ordinary skill in the art at the time of the invention as reflected in the prior art in this proceeding, although we agree with Patent Owner that a degree in plant physiology, horticulture, plant science or the like along with work experience, or a master’s degree in exchange for such work experience, in the area of controlled release of compounds would provide sufficient expertise to understand the teachings of the ’185 patent in a manner necessary to apply them.

Accordingly, we adopt the following definition of the skilled artisan for purposes of this Decision:

a bachelor’s degree in chemistry, chemical engineering, materials science, plant physiology, horticulture, plant science or the like, and 1–2 years of work experience, or a master’s degree in exchange for such work experience, in the area of controlled release of compounds, such as those used to inhibit ethylene response in plants.

C. Claim Interpretation

We apply the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 42.100(b) (2020). Under that standard, claim terms “are generally given their ordinary and customary meaning” as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Extrinsic evidence is “less significant than the intrinsic record in determining ‘the legally operative meaning of claim language.’” *Phillips*, 415 F.3d at 1317.

Neither party advances any claim terms for construction. Pet. 20; PO Resp. 15. We find that no express construction is necessary for purposes of rendering this Decision and apply the plain meaning of all terms. *See Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011) (“[C]laim terms need only be construed ‘to the extent necessary to resolve the controversy.’”) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

D. Overview of the Prior Art

1. Summary of Hisano (Ex. 1004)

Hisano is the translation of a Japanese patent application published on July 5, 1994. Hisano teaches an “effervescent tablet preparation for keeping cut flower freshness, which is characterized in that it contains a component for keeping cut flower freshness and an effervescence component comprised

of carbonate and water-soluble solid acid.” Ex. 1004 ¶ 9. Hisano discloses that a test of cut flower freshness showed that the composition “was either noninferior or superior in terms of freshness preserving benefits, there was no liquid cloudiness or harmful chemical effects observed, and it demonstrated superior benefits as a cut flower freshness preserving agent.” *Id.* ¶ 24.

The tablet formation includes: 1) a “non-chlorine-based component” to preserve freshness that can include “silver thiosulfate . . . [or] ethylene suppression agents (such as aminooxyacetic acid)” and 2) “carbonate and water-soluble acid as the effervescence component.” *Id.* ¶¶ 10, 13. Additional components “such as aromatic components, pigments, excipients, etc.” can be added to the formulation. *Id.* ¶ 16. Hisano teaches methods for manufacturing the disclosed tablet. *Id.* ¶¶ 14–16. Hisano discloses the following regarding use of the tablet: By inserting the effervescent tablet preparation for keeping cut flower freshness described by the present invention comprised of the aforementioned compositions into water located in flower containers, dissolution will begin, neutralizing reactions will occur between the carbonate and the water-soluble solid acid included therein, where after the freshness preserving components will disperse throughout the solution in the flower container, and the generated carbonic acid gas will promote assimilation action, which will be effective for cultivating, and preserving the freshness of, the cut flowers (plants). *Id.* ¶ 25.

2. *Summary of Daly (Ex. 1005)*

Daly issued January 25, 2000, and discloses methods for inhibiting the ethylene response in plants. Ex. 1005, code (57). Daly discloses that one method of inhibition is use of ethylene synthesis inhibitors, which “reduce the quantity of ethylene that a plant can produce,” but that these inhibitors

do not suppress the negative effects of ethylene from environmental sources. *Id.* at 2:20–33. An alternative method is to block the receptor site that signals ethylene action, through use of a compound such as silver thiosulfate (STS). *Id.* at 2:42–48. However, STS has a “serious waste disposal problem” and is almost exclusively used by growers. *Id.* at 2:54–62. Consequently, “there is a great desire . . . to find alternatives to STS [with the] only commercially acceptable replacements for STS [being] cyclopropene, cyclopentadiene, diazocyclopentadiene and their derivatives.” *Id.* at 2:63–67.

Daly discloses that an “effective agent for inhibiting the ethylene response in plants should provide an irreversible blocking of the ethylene binding sites and thereby allow treatments to be of short duration” rather than requiring continuous or repeated treatments. *Id.* at 3:4–11. Daly teaches that cyclopropene and its derivatives, including methylcyclopropene, are effective blocking agents for ethylene binding, do not have the odor problem of other such agents, but that they are “relatively unstable due to their potential for undergoing oxidation and other reactions,” leading to “a problem of stability of these gases, as well as the explosive hazards these gases present when compressed.” *Id.* at 3:20–29. In particular, Daly notes that “strained carbenes such as methylcyclopropene are quite unstable to reaction with oxygen, self polymerization and reaction with other organic compounds.” *Id.* at 10:40–44.

To solve this issue, Daly discloses

a method of incorporating these gaseous compounds, which inhibit the ethylene response in plants, in a molecular encapsulation agent complex in order to stabilize their reactivity and thereby provide a convenient and safe means of storing, transporting and applying or delivering the active compounds to

plants. The application or delivery methods of these active compounds can be accomplished by simply adding water to the molecular encapsulation agent complex.

Id. at 3:30–47. To address the stability and potential explosiveness issues, Daly discloses “a molecular encapsulation agent complex that stabilizes the reactivity of these gases and thereby provides a convenient and safe means of storing, transporting and applying or delivering these gases to plants.” *Id.* at 3:43–47. When in the complex, the method of delivery “comprises bubbling a gas through a solvent while it is in contact with the complex.” *Id.* at 7:53–55.

Daly discloses in Example 3 (*id.* at 17:59–18:31) an exemplary release of methylcyclopropene from cyclodextrin molecular encapsulation using water, and provides recommendations and parameters to achieve effective release to provide ethylene receptor blocking treatment. Using this method, Daly states:

Once the plants are treated, the sealed chamber may be opened if desired. The methylcyclopropene is now protecting the plants because it has blocked all the available ethylene receptor sites. This treatment will protect the plants from the action of ethylene until the plant grows new unblocked ethylene receptor sites.

Id. at 18:26–31.

E. Ground 2 – Obviousness over Hisano and Daly

1. Petitioner’s Contentions

Petitioner contends that claims 3 and 11 of the ’185 patent are unpatentable under 35 U.S.C. § 103(a) as obvious over Hisano in view of Daly. Pet. 36–53. Because claims 3 depends from claim 2, which in turn depends from claim 1, and because claim 11 also depends from claim 1, Petitioner incorporates its allegations that Hisano anticipates claim 1, which

we review below in conjunction with the additional limitations added by dependent claims 2 and 3. *See* Pet. 37 (incorporating allegations set forth at Pet. 28–31).

a) Claim 3

(1) An effervescent tablet dosage comprising:

Petitioner argues that Hisano discloses “[a]n effervescent tablet preparation for keeping cut flower freshness, which is characterized in that it contains a component for keeping cut flower freshness and an effervescence component comprised of carbonate and water-soluble solid acid.” Pet. 36 (citing Ex. 1004 ¶¶ 9, 29; Ex. 1006 ¶ 68). Petitioner argues that the skilled artisan would have understood that Hisano’s tablet is a dosage as recited in the claims because Hisano discloses that the user would find it possible to “appropriately select the mixing ratio” while maximizing benefits and avoiding danger to the treated plants as well as considering the time the tablet would take to dissolve, the necessary effervescence effects, and the quality of the resulting product. *Id.* (citing Ex. 1004 ¶¶ 14, 19).

(2) an agent for blocking the ethylene binding site in plants, wherein the agent for blocking the ethylene binding site in plants is selected from the group consisting of cyclopropene, 1-methylcyclopropene, 3,3-dimethylcyclopropene, methylenecyclopropane, diazocyclopentadiene, transcyclooctene, cis-cyclooctene, and 2,5-norbornadiene, the derivatives thereof, and the mixtures thereof.

Petitioner argues that Hisano discloses inclusion of an agent such as STS known to the skilled artisan as effective in blocking the ethylene binding site as a component to keep cut flowers fresh. *Id.* at 29 (citing Ex. 1004 ¶¶ 8, 13; Ex. 1006 ¶ 69; Ex. 1005, 2:42–52; Ex. 1010, 201).

Petitioner argues that Daly teaches that the molecular encapsulation of cyclopropene and its derivatives, including 1-methylcyclopropene, provides an irreversible blocking of the ethylene binding sites in plants, and that the skilled artisan would have understood the molecularly encapsulated version of 1-MCP to be an agent for blocking the ethylene binding site in plants. Pet. 38 (citing Ex. 1005, 3:1–48; Ex. 1006 ¶ 92).

(3) an effervescent ingredient

Petitioner argues that the tablet disclosed in Hisano teaches this limitation by providing an “effervescence component” that includes carbonate and water-soluble acid. Pet. 30 (citing Ex. 1004 ¶¶ 8, 10; Ex. 1006 ¶ 72). Petitioner argues that Hisano discloses sodium carbonate, malic acid, and citric acid as examples of water-soluble acids that would have been known to the skilled artisan. *Id.* (citing Ex. 1004 ¶ 11; Ex. 1006 ¶ 72).

(4) in admixture with one or more acceptable carriers and/or excipients

Petitioner argues that Hisano teaches use of “other components” in the tablet including “excipients” and discloses saccharose, which is a common acceptable binder for tablet formulations. *Id.* at 30–31 (citing Ex. 1004 ¶ 19). Petitioner argues that the skilled artisan would have understood Hisano’s general disclosure of use of an excipient, and specific disclosure of saccharose to teach that conventionally used excipients could be used in the formulation. *Id.* at 31 (citing Ex. 1006 ¶ 73). Petitioner argues that the skilled artisan would have understood that “in admixture” means “mixed with” and that Hisano teaches mixing the formula components, such as the mixture of “saccharose excipient with a freshness preserving agent and effervescent ingredient” as shown in Table 1. *Id.* (citing Ex. 1004 ¶ 18).

b) Claim 11

Claim 11 requires that “the agent for blocking the ethylene binding site in plants is released in gaseous form.” Ex. 1001, 6:49– 50.

Petitioner argues that Daly “teaches that, when the complex containing the encapsulated 1-MCP is dissolved in a solvent, such as water, the 1-MCP gas is released and the treated plant will be exposed to the gas.” Pet. 40 (citing Ex. 1005, 7:25–59, 17:48–18:31). Petitioner argues that the skilled artisan would have understood that the active ingredient in Daly, 1- MCP, would be released from the complex in gaseous form. *Id.* (citing Ex. 1006 ¶¶ 99–100).

Based on the cited teachings as set forth above, we find that Petitioner has sufficiently shown that each of the limitations of claims 3 and 11 is taught by the disclosures of Hisano and Daly. Patent Owner does not dispute that the combination of Hisano and Daly teaches the claim limitations recited in claims 3 and 11. *See generally* PO Resp.

2. Motivation to Combine Hisano and Daly

Petitioner argues that a skilled artisan would have been motivated to combine the teachings of Hisano and Daly for three reasons. Pet. 42–46. First, Petitioner argues that the skilled artisan would have understood that tablets were desirable forms for delivering dosage of an active ingredient for suppressing or blocking ethylene production in a plant. *Id.* at 42–43. Petitioner, based both on prior art and testimony of its expert, argues that tablets were desirable as they “allowed for the delivery [of] an accurate dosage of an active ingredient, avoided the need to weigh, mix or measure associated with other dosage forms, such as liquids or powders, and could be more easily packaged and transported.” *Id.* (citing Ex. 1004 ¶¶ 3–5, 7; Ex. 1018, 1:16–30; Ex. 1006 ¶ 110). Petitioner argues that the skilled

artisan would have been aware of the “benefits of using an effervescent tablet preparation to deliver to plants an ethylene blocking agent, such as STS, and/or an ethylene suppression agent, such [as] AOA [aminooxyacetic acid],” which included improved control over the release of active ingredients through dissolution and dispersion. *Id.* at 43 (citing Ex. 1004 ¶¶ 7, 12, 14, 15, 25–28; Ex. 1016, 1:8–15; Ex. 1019, 1:31–42, 2:35–52; Ex. 1006 ¶ 110).

Petitioner further argues that the skilled artisan would have understood that the ethylene suppression agent disclosed in Hisano only inhibited a plant’s ability to produce ethylene and did not combat environmental exposure sources. *Id.* at 43. Petitioner contends that the skilled artisan would have found STS an attractive option because its use would permanently block ethylene binding sites from any source, not just internally produced ethylene. *Id.* at 43–44 (citing Ex. 1005, 2:42–47; Ex. 1010, 201; Ex. 1006 ¶ 111). Petitioner argues that the artisan would further have understood that disposing of STS was problematic due to its silver content, and that use of a substitute without a silver byproduct requiring hazardous waste disposal would be advantageous. *Id.* at 44 (citing Ex. 1005, 2:54–63; Ex. 1006 ¶ 112). For these reasons, Petitioner argues, a skilled artisan would have been motivated to find a substitute for STS that had a similar ability to permanently block ethylene binding sites, but was also easier to dispose of. Thus, a skilled artisan would have been motivated by Daly’s disclosure that “cyclopropene, cyclopentadiene, diazocyclopentadiene and their derivatives [were] ‘the only commercial acceptable replacements for STS’ at the time” to explore use of those compounds as suitable alternatives to STS. *Id.* (citing Ex. 1005, 2:62–67; Ex. 1006 ¶ 112).

Petitioner next argues that the skilled artisan would have found the “combination of Daly’s 1-MCP powder complex with Hisano’s effervescent tablet to amount to a substitution of one known element for another to yield predictable results” because 1-MCP was to be used for the same purpose as Daly’s agent and each is powder-based. *Id.* at 43–44 (citing Ex. 1001, 1:51–56; Ex. 1004 ¶¶ 15, 18; Ex. 1005, 2:20–67, 11:1–13; Ex. 1006 ¶ 113).

Petitioner argues that because the methods for making the compositions were largely the same, the skilled artisan would have believed Hisano’s ethylene suppression agent or STS could be substituted with Daly’s 1-MCP in powder form and “used to form an effervescent tablet in any of the ways well known in the art,” including the method disclosed in Hisano. *Id.* at 45–46 (citing Ex. 1017, 1:23–26; Ex. 1018, 1:63–75, 2:33–3:5; Ex. 1006 ¶ 114; Ex. 1004 ¶¶ 14, 15, 18). Petitioner argues that the interchangeability of the components is confirmed by the acknowledgment in the ’185 patent that the skilled artisan could practice the invention of the patent using “conventional” tablet dosage form preparation processes. *Id.* at 46 (citing Ex. 1001, 4:28–37; Ex. 1006, 114).

Finally, Petitioner argues that 1-MCP as taught by Daly was “among a finite number of known compounds” that could be substituted for Hisano’s STS compound or for AOA to provide “safer, effective ethylene blocking,” and that it would have been obvious for the skilled artisan to have selected 1-MCP for substitution “to provide a more consumer-friendly, safer and more effective treatment for prolonging the shelf life of plants, including cut flowers.” *Id.* (citing Ex. 1006 ¶ 115; Ex. 1005, 2:17–3:11; Ex. 1004 ¶¶ 2–7; Ex. 1013, 11).

After having reviewed the record and the arguments of the parties, we find that the weight of the evidence supports Petitioner’s allegation that one

of skill in the art at the time of the invention would have been motivated to combine the teachings of Hisano and Daly to make the subject matter of claims 3 and 11. We agree that 1-MCP, as among a finite number of known compounds, would have been the substitution of one known element, STS, for another, 1-MCP, that yields predictable results. We address Patent Owner's arguments challenging Petitioner's proffered motivation to combine below.

Patent Owner argues that Petitioner fails to meet its burden to show that the skilled artisan would have been motivated to combine the teachings of Hisano and Daly to make the claimed subject matter. PO Resp. 22–41. First, Patent Owner argues that the skilled artisan would not have been motivated to incorporate a gaseous agent into Hisano's tablet. *Id.* at 22. Patent Owner, through its expert Dr. Beaudry, argues that Hisano does not contemplate inclusion of a gaseous agent within the tablet and that doing so would have been “at odds with the purpose of Hisano's alleged invention—to activate the freshness-preserving agent in a localized manner to keep cut flowers fresh longer.” *Id.* (citing Ex. 2013 ¶¶ 44, 51–53). Patent Owner argues that Hisano aimed to solve “problems associated with evenly dispensing a liquid or solid freshness preservation agent locally within the confines of the water within a flower vase to keep the exposed cut flowers fresh for longer.” *Id.* (citing Ex. 2013 ¶¶ 52–53). *See also* PO Sur-reply, 13–15.

Patent Owner argues that every freshness preserving agent disclosed in Hisano is a local-acting solid or liquid, not a gas. *Id.* at 23–24 (citing Ex. 1004 ¶ 13; Ex. 2010, 10, 13, 21, 29–30; Ex. 1009, 278–279; Ex. 1014, 1:9–59, 2:80–88; Ex. 2013 ¶¶ 42–43). Patent Owner argues that the skilled artisan would have known that 1-MCP and other potential gaseous active

compounds function by release into the atmosphere in a sealed environment for dispersal to all available ethylene sites. *Id.* at 25 (citing Ex. 2013 ¶¶ 53–54). Patent Owner argues that 1-MCP had not been “approved for direct contact with plants,” as of 2000, and was only approved for separate solution-based aerosol application to treat plants on a large scale. *Id.* at 26 (citing Ex. 2013 ¶ 54). For these reasons, Patent Owner argues, a skilled artisan would not have believed that 1-MCP and other potential gaseous active compounds would be suitable for use to preserve freshness on a small scale such as with cut flowers. *Id.* (citing Ex. 2013 ¶¶ 52–54). Dr. Beaudry testifies that the skilled artisan would have understood that use of 1-MCP on the small scale of a flower vase “would have been inefficient, to the extent it worked at all.” Ex. 2013 ¶ 53.

Patent Owner further argues that Petitioner’s predecessor’s seeking of a patent for encapsulation of 1-MCP gas into an effervescent tablet at the same time as Patent Owner is contemporaneous evidence of non-obviousness. PO Sur-reply, 15 (citing Ex. 2002 at claim 9, 3:38–4:4, 4:49–50).

We are not persuaded. Patent Owner cites no case law requiring us to hold the actions of a predecessor of a petitioner in interest against the petitioner, and we decline to do so or to find that the predecessor’s application for and obtaining of a patent on similar technology is evidence of nonobviousness in a separate proceeding.

With regard to the evidence Petitioner cites, Dr. Beaudry’s testimony on the issue of motivation to combine, while accurate and helpful in light of his experience in the field, focuses on the limitations of the individual disclosures of Hisano and Daly without consideration of the inferences that the skilled artisan would have drawn from the totality of the teachings of the

art at the time of the invention, as is necessary in an obviousness challenge based on the teachings of multiple references. “The test of obviousness *vel non* is statutory. It requires that one compare the claim’s ‘subject matter as a whole’ with the prior art ‘to which said subject matter pertains.’” *In re Ochiai*, 71 F.3d 1565, 1569 (Fed. Cir. 1995) (quoting 35 U.S.C. § 103).

Claims 3 and 11 recite a tablet dosage containing an effervescent ingredient and an agent “for blocking the ethylene binding site in plants,” but require no effectiveness level or other measure of the blocking action to be achieved. Ex. 1001, 6:11–49. Although Hisano’s examples focus on delivery within a flower container, Hisano makes no statements restricting its use to cut flowers and suggests broader application including use of alternative ethylene suppression agents including STS with effervescent tablets. Ex. 1004 ¶ 13. One of ordinary skill in the art can learn from what any reference teaches or suggests and apply it for the artisan’s own purpose. *See In re Mouttet*, 686 F.3d 1322, 1331 (Fed. Cir. 2012) (“A reference may be read for all that it teaches, including uses beyond its primary purpose”). Moreover, patent law does not require that the teachings of the reference be combined for the reason or advantage contemplated by the inventor, as long as some suggestion to combine the elements is provided by the prior art as a whole. *See KSR*, 550 U.S. at 419–20:

In determining whether the subject matter of a patent claim is obvious, neither the particular motivation nor the avowed purpose of the patentee controls. . . . [A]ny need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed.

We find that the skilled artisan, as we have defined herein to be a person with work experience in the area of controlled release of compounds

such as those used to inhibit ethylene response in plants, would have looked at Hisano as part of the artisan's consideration of analogous art. *See, e.g., Princeton Biochemicals, Inc. v. Beckman Coulter, Inc.*, 411 F.3d 1332, 1339 (Fed. Cir. 2005); *In re GPAC Inc.*, 57 F.3d 1573, 1577–78 (Fed. Cir. 1995). (“The scope of the prior art includes all analogous art.”) Hisano is in the same field of endeavor – blocking ethylene receptors to regulate plant physiology. Even if we were to consider Hisano's smaller scale a factor in determining whether it is analogous art, we find that Hisano's disclosures regarding blocking ethylene receptors to prolong plant life are directly pertinent to the very same problem addressed by the '185 patent inventors. *Princeton Biochemicals*, 411 F.3d at 1339; *In re GPAC Inc.*, 57 F.3d at 1577–78) (defining test for analogous references).

We find that the ordinarily skilled artisan would have looked at Hisano as an analogous reference and would have been motivated to try scaling up its smaller-scale solution of effervescent tablets for delivery on a larger scale than single flower vases. We further find that the artisan would not have been deterred by Hisano's lack of embodiments including gas agents and disclosing local use only that would have prevented use of 1-MCP directly with plants. Ex. 2013 ¶¶ 52–54. In particular, because Daly suggested the molecularly encapsulated 1-MCP could be made in powder form (Ex. 1005, 11:1–11), which is the form Petitioner suggested for use in the proposed obviousness combination (e.g., Pet 36–37), we are not persuaded by Patent Owner's arguments regarding concerns the skilled artisan would have had about working with 1-MCP gas that is released into the atmosphere where the atmosphere becomes the treatment medium as opposed to a liquid medium.

We find Hisano's direct mention of STS as a potential alternative agent, along with Daly's teachings that 1-MCP is a known substitute for STS without STS's toxic waste byproduct would have motivated the artisan to explore use of 1-MCP. Daly's teaching of using a cyclodextrin cage-stabilized and powdered version of 1-MCP to inhibit ethylene binding in plants would have motivated the skilled artisan to try this stabilized version in Hisano's effervescent tablet preparation because of the benefits of effervescent ingredients for dissolution and dispersion, to treat plants on a larger scale. *See* Ex. 1005 11:1–20; Ex. 1006 ¶¶ 29, 110–115; Ex. 1035 ¶¶ 11–18. Hisano instructs the artisan about optimization by creating tablets with a proper “effervescence component: freshness preserving component to a weight ratio of between 6.5:3.5 and 5.0:5.0 to create a large, compact tablet agent that can secure some degree of the freshness preserving component” while also stabilizing the tablet for transportation and adjusting the ratios of components to affect the “target quality, effervescence effects, and dissolution time, etc. of the final product.” Ex. 1004 ¶¶ 12, 14. These additional teachings would have led the skilled artisan to understand that an effervescent tablet dosage made with cyclodextrin-stabilized 1-MCP could be made for larger scale production and use.

In particular, we find that the artisan would have been motivated to combine the teachings of Hisano and Daly because of 1) Daly's teaching that STS is problematic for use in inhibiting ethylene in plants due to difficulties in silver waste byproduct disposal, leading to “a great desire among postharvest physiologists to find alternatives to STS” (Ex. 1005, 2:42–64); 2) Daly's teachings that cyclopropene and its derivatives, including 1-MCP, are “commercially acceptable replacement[s] for STS”) (*id.* at 1005) and that powder complexes provided advantages over

compressed gases for safety and control (Ex. 1005, 2:44–68; 4:13–24); and 3) Hisano’s teaching that an ethylene suppressing agent could be delivered via an effervescent tablet, which when dissolved in water, creates “generated carbonic acid gas [that] will promote assimilation action, which will be effective for cultivating, and preserving the freshness of, the cut flowers (plants).” Ex. 1004 ¶ 25; *see also* Ex. 1006 ¶¶ 113–115 (opining that the skilled artisan would have understood encapsulated 1-MCP, a powder-based form, to be one of a finite number of safer substitutes for STS that could be incorporated into an effervescent tablet using methods known to the skilled artisan at the time); Ex. 1013 at 9 (identifying 1-MCP as a newly discovered “ethylene inhibitor [that acts] as a binding site competitor and is very promising as a postharvest treatment for both potted plants and fresh cut flowers [giving] ethylene protection equal to that obtained from STS, but without silver” and noting that a “company plans to sell the product in a powder form that will release MCP gas when mixed with water”). The art suggested finding substitutes for STS, meaning that the skilled artisan would have looked for likely substitutes such as this stable form of 1-MCP for use in various delivery methods. *See, e.g., In re Kerkhoven*, 626 F.2d 846, 850 (“It is *prima facie* obvious to combine two compositions each of which is taught by the prior art to be useful for the same purpose, in order to form a third composition which is to be used for the very same purpose.”).

Because of this evidence, Dr. Beaudry’s testimony does not persuade us that the skilled artisan would not have believed that cyclodextrin-caged 1-MCP could overcome the concerns Dr. Beaudry identifies. Ex. 2013 ¶¶ 52–54. Notably, Daly’s teaching that 1-MCP could be in a stable cyclodextrin cage and powdered form allays any concerns regarding gas instability, including the fact that 1-MCP gas had not been approved for direct contact

with plants. Ex. 1001, 1:51–56. Patent Owner has not adequately rebutted Petitioner’s evidence on this issue by demonstrating why a gas in the form of cyclodextrin-caged, powdered product such as 1-MCP could not have been used.

The 1-MCP gas described in Patent Owner’s cited references, *see e.g.*, Ex. 2017, was not present in a cyclodextrin cage, and thus could only be provided in higher concentrations in a sealed environment that would allow controlled delivery. Ex. 2017, 1. Caging the 1-MCP gas as disclosed in Daly permitted stability and release of the 1-MCP by delivery by contacting the 1-MCP complex with a solvent. Ex. 1005, 11:4–11; 7:50–52. Daly also discloses that “[a] more specific feature of this cyclopropene method of delivery embodiment comprises bubbling a gas through the solvent while it is in contact with the complex.” *Id.* at 7:53–55. We are persuaded that the skilled artisan would have viewed this disclosure in Daly as an invitation to try use of an effervescent compound as taught by Hisano, which discloses:

Action / Effects of the Invention] By inserting the effervescent tablet preparation for keeping cut flower freshness described by the present invention comprised of the aforementioned compositions into water located in flower containers, dissolution will begin, neutralizing reactions will occur between the carbonate and the water soluble solid acid included therein, where after the freshness preserving components will disperse throughout the solution in the flower container, and the generated carbonic acid gas will promote assimilation action, which will be effective for cultivating, and preserving the freshness of, the cut flowers (plants).

Ex. 1004 ¶ 25.

We agree that the artisan, having read the teachings of Hisano and Daly, would have been motivated to use stable, powdered 1-MCP in an effervescent tablet as a means to effectively disperse the gas when subjected

to water. We are persuaded by Dr. Walton's testimony that the skilled artisan would have believed that Daly's 1-MCP powder complex could have been mixed with Hisano's effervescent ingredient and excipient and pressed into a tablet to be released by exposure to water, and would have been motivated to make the combination due to the ease of delivery and precision that tablet dosages provide. Ex. 1006 ¶¶ 109–118.

3. *Reasonable Expectation of Success in Releasing 1-MCP from Hisano's Tablet*

Petitioner alleges that a skilled artisan at the time of the invention would have had a reasonable expectation of success in combining the teachings of Hisano and Daly. Pet. 47–53. Petitioner, through Dr. Walton's testimony, argues that the artisan would have known that conventional processes for preparing dosage formations in tablets, as disclosed in the '185 patent specification, could be used to prepare a formulation incorporating Daly's 1-MCP powder into Hisano's tablet formulation. *Id.* at 47 (citing Ex. 1006 ¶¶ 116, 117). Petitioner argues that the artisan would have believed that the powder complex would remain stable, with the 1-MCP gas remaining contained in the cyclodextrin complex so long as the complex was not contacted with a solvent. *Id.* at 48 (citing Ex. 1006 ¶ 118). *See also* Ex. 1035 ¶ 14. Dr. Walton cites Sisler⁶ for its disclosure that 1-MCP can withstand incorporation with adjuvants or carriers and tumbling with a solid carrier, meaning that "physical mixing should not pose any particular hazard." *Id.* at 48–49 (citing Ex. 1012, 4:56–57, 5:14–20, 5:56–62; Ex. 1006 ¶ 119). Dr. Walton opines that the skilled artisan would have understood that the cyclodextrin-caged 1-MCP protects against gas release unless

⁶ Edward C. Sister and Syliva M. Blankenship, U.S. Pat. No. 5,518,988, issued May 21, 1996 (Ex. 1012, "Sisler").

disrupted by a solvent, and can withstand mechanical forces needed to create a tablet such that a solvent is required to release the gas. *Id.* at 49–50 (citing Ex. 1022, 19–20; Ex. 1006 ¶¶ 121, 122, 126; Ex. 1005, 12:12–51, 17:61–18:31). Dr. Walton opines that, to the extent the skilled artisan had concerns that the 1-MCP gas would not reliably release from the tablet, the skilled artisan would have been able to adjust the tablet compression to “vary the tablet’s hardness, friability, impact and shock resistance, speed of disintegration and delivery of active ingredients”. *Id.* at 51 (citing Ex. 1004 ¶ 15; Ex. 1015, 1:51–66, 6:29–56; Ex. 1006 ¶ 123).

Dr. Walton opines that Daly’s disclosure provides sufficient information for the skilled artisan to have believed that “1-MCP gas would not be released from an alpha-cyclodextrin encapsulation under lower pressures, particularly because 1-MCP powder placed in an aerosol can under “40-50 psi [e.g., approximately 3-3.5 kg/cm²] of compressed gas” for aerosol delivery still required “sufficient water” to release the 1-MCP gas. *Id.* at 52 (citing Ex. 1005, 17:55–58; Ex. 1006 ¶ 124). Dr. Walton opines that the skilled artisan would have reasonably believed that making “an effervescent tablet using complexed 1-MCP” would have been successful. *Id.* at 52–53 (citing Ex. 1006 ¶ 125). Petitioner adds that the ’185 patent does not mention specific concerns with tableting or provide teachings to ensure successful tableting, but rather states that “all conventional processes” for preparing tablet dosages could be used. *Id.* at 52–53 (citing Ex. 1001, 4:28–37, 5:10–19, 6:11–14, 6:51–55; Ex. 1006 ¶ 125). Petitioner argues that, based on the above information, the skilled artisan would have understood that 1-MCP required a liquid solvent “as well as potentially high temperature” to break the molecular

encapsulation interactions and release the 1-MCP from the alpha-cyclodextrin cavity,” and thus a skilled artisan would have reasonably expected that Daly’s 1-MCP complex could be mixed with an effervescent ingredient as taught in Hisano and compressed into a tablet without release of the 1-MCP gas. *Id.* at 53 (citing Ex. 1006 ¶ 126).

After having reviewed the record and the arguments of the parties, we find that the weight of the evidence supports Petitioner’s conclusion that one of skill in the art at the time of the invention would have reasonably expected success in making the claimed subject matter for the reasons set forth above. We address Patent Owner’s arguments against a reasonable expectation of success below.

Patent Owner argues that the ordinary artisan would not have been motivated to incorporate Daly’s 1-MCP into Hisano’s tablet or have had a reasonable expectation of success in the endeavor. PO Resp. 29–41. Patent Owner argues that the artisan would not have made the claimed combination having believed that a “gaseous active agent—whether molecularly encapsulated or otherwise—would have diffused into the atmosphere upon release and would not have achieved Hisano’s purpose of preserving the freshness of the cut plant.” *Id.* at 30 (citing Ex. 2013 ¶ 53; Ex. 2018, 28:15–20, 30:7–14). Patent Owner argues that because neither Hisano or Daly discloses “incorporating encapsulated 1-MCP into a tablet, let alone successfully releasing 1-MCP from such a tablet,” the artisan would not have understood the “unique challenges” that releasing a gas would have created, which “would have been further compounded by tableting.” *Id.* at 29–30 (citing Ex. 2013, ¶¶ 23–26, 52, 55, 58–59, 60–64).

Patent Owner argues that “molecularly encapsulated 1-MCP simply does not behave like a conventional solid active agent” and the skilled artisan would have known that “release of 1-MCP from Daly’s α -cyclodextrin complex was not accomplished simply by adding Daly’s complex to water and dissolving it,” because 1-MCP release slows when 1-MCP is released into solution, leaving empty cyclodextrin molecules that “prevent the release of additional 1-MCP gas molecules, precluding full release.” *Id.* at 29, 32 (citing Ex. 2013 ¶¶ 57–59, 64). Patent Owner further argues that Petitioner’s predecessor argued that the formulation was surprisingly efficient and therefore not obvious. *Id.* at 32–33 (citing Ex. 2002, 2:16–18). Patent Owner concludes the skilled artisan would not have believed the substitution of STS or AOA with 1-MCP would reasonably result in delivery of 1-MCP gas. *Id.* at 33 (citing Ex. 2013 ¶¶ 58–61).

We are not persuaded.⁷ The referenced testimony from Dr. Walton’s deposition from a separate litigation relates to whether the release of a gaseous active agent escaping from water must be in a sealed container in order to reach a target concentration. Ex. 2018, 28:15–20, 30:7–14. While Dr. Walton agrees that a sealed container would be necessary for a target concentration of active ingredient to be reached, she also testifies that “if there’s even one gas agent (ethylene blocker) that can touch the plant, it will block that site,” and that the sealed container would be used to increase the efficiency of reaching the target concentration, not to achieve blocking. *Id.* at 29:11–17.

⁷ Patent Owner recites in this discussion the same arguments as to why the artisan would not have been motivated to make the claimed combination (e.g., PO Resp. 30), which we have addressed above and do not repeat here.

Dr. Walton’s testimony in that separate matter is not persuasive as to the subject matter of Challenged Claims 3 and 11 in this proceeding. “The reasonable-expectation-of-success analysis must be tied to the scope of the claimed invention.” *Teva Pharmaceuticals USA, Inc. v. Corcept Therapeutics, Inc.*, 18 F.4th 1377, 1381 (Fed. Cir. 2021). Neither claim 3 nor 11 recites any required level of ethylene site blocking. Because no effectiveness or release level is recited, we are not persuaded that the skilled artisan would have believed creating the recited dosage form would have required a closed chamber. For this same reason we are unpersuaded that an inability to achieve “full release” would have led the skilled artisan to believe success could not be achieved. Nor are we persuaded that the lack of prior teaching of tablets containing encapsulated 1-MCP is necessary for a reasonable expectation of success. Daly’s teachings regarding the stability of the encapsulated 1-MCP absent direct exposure to a solvent (*see* Ex. 1006 ¶ 126) suggest the artisan would have reasonably believed the stable 1-MCP could be incorporated into a tablet, and Daly’s teachings that 1-MCP gas in compressed tablet form could be released by dissolution with water or by bubbling a gas through a solvent that is in contact with the complex (Ex. 1005, 7:53–55) are strong evidence that the artisan would have reasonably believed success could be achieved.

Patent Owner next argues that Daly does not suggest that tableting would be successful and that the skilled artisan would have understood that tableting Daly’s 1-MCP molecular encapsulation complex “would have made releasing the 1-MCP gas from the cyclodextrin carrier molecules even more challenging.” *Id.* at 33 (citing Ex. 2013 ¶¶ 60–61). Patent Owner argues that Daly is silent on whether 1-MCP could be incorporated into a tablet and, once compressed, effectively release 1-MCP gas, and that

Dr. Walton does not address whether the artisan “would have had a reasonable expectation of successfully releasing the 1-MCP gaseous agent from its cyclodextrin encapsulation agent within the tablet.” *Id.* at 33–34 (citing Ex. 2019 ¶ 77; Ex. 2020 ¶ 56). Patent Owner argues, through Dr. Beaudry, that “release of the gaseous molecules from the cyclodextrin complexes was not straightforward, as it required a particular volume of solution and release kinetics to successfully solubilize and release.” *Id.* at 34 (citing Ex. 2013 ¶ 59). Patent Owner cites references regarding the release of 1-MCP from an encapsulation agent, which explain that the release requires an excess of water and can be affected by other present compounds. From this, Patent Owner concludes that the skilled artisan would not have had a reasonable expectation of success.

We are not persuaded. The testimony excerpts Patent Owner cites from Dr. Walton relate to whether 1-MCP was a well-known gas as of the time of application for a patent in a different lawsuit with different claims. The issue here relates to whether the skilled artisan, having read the cited articles in the prior art, would have believed the claimed subject matter was obvious. Patent Owner has not argued or otherwise shown that the cited references in that matter would not have been considered by one of skill in the art as defined herein; thus, we find these unrelated statements not persuasive for the issue at hand. Patent Owner’s citation of brief testimony from that lawsuit is out of context and does not sufficiently show that the statements have any bearing on the issue in this case.

Regarding the release of gaseous molecules from the cyclodextrin complexes, Patent Owner and Dr. Beaudry have not addressed what “an excess” of water would be and to what extent other compounds would influence solubility. Without that information, these statements are

unpersuasive with regard to claims 3 and 11, which do not recite any solubility or treatment effectiveness parameters, particularly in light of Daly's teachings that the method of delivery for encapsulated 1-MCP "comprises bubbling a gas through the solvent while it is in contact with the complex . . . or applying heat to the solvent either before it contacts the complex or during that contact." Ex. 1005, 8:12–19. Patent Owner does not persuade us that, in light of this teaching, the skilled artisan would have understood that a tablet containing 1-MCP could not be induced to release the 1-MCP by simple dissolution in warm water.

Patent Owner next argues that Petitioner's arguments as to why encapsulated 1-MCP would be stable are irrelevant because they have nothing to do with compression or tableting, but rather the complex's ability to withstand ultraviolet degradation and heating. PO Resp. 35–36 (citing Pet. 47–61; Ex. 1005, 4:50–52; Ex. 2013 ¶ 63).

We are not persuaded. Daly's teachings include that cyclodextrin-caged 1-MCP powder can be released from the complex by exposure to a solvent or by heating. Ex. 1005, 8:12–19; *see also* Reply 19. Increased resistance to release due to heat in the cyclodextrin-caged stage is thus some evidence of increased stability. The evidence of record also shows that temperature changes during tableting were minor, and that the artisan would have known how to mitigate temperature changes to assist the tableting process, resulting in minimal if any loss of material. Ex. 1035 ¶ 43–44; Ex. 1028, 1524–25; Ex. 1029, 186–88; Ex. 1030, 527–28; Ex. 1006 ¶ 123; Ex. 1035 ¶¶ 43–44. Moreover, the '185 patent's statement that "[p]ersons skilled in the art of preparing tablet dosage formulations may adopt all the conventional processes to practice the tablet dosage of the present invention" is strong evidence to the contrary of Patent Owner's assertions concerning

what one of skill in the art would have understood about the stability of 1-MCP. *See Qualcomm Inc. v. Apple Inc.*, 24 F.4th 1367, 1376 (Fed. Cir. 2022) (finding that admissions are “permissible evidence in an *inter partes* review for establishing the background knowledge possessed by a person of ordinary skill in the art” and “provide a factual foundation as to what a skilled artisan would have known at the time of invention”).

4. *Reliance on Alleged Out-of-Context Statements*

Patent Owner argues that Petitioner’s reliance on certain references for support is misconstrued. PO Resp. 37–41. Patent Owner alleges that Sisler ’988 (Ex. 1012), which discloses “tumbling” an active ingredient [cyclopropene] with a solid carrier, does not support the notion that the ingredient could be used to create a solid tablet, which requires higher compression forces. *Id.* at 37–41 (citing Ex. 2013 ¶ 66).

Patent Owner alleges that Szejtli (Ex. 1022), which discloses that complexed compounds not exposed to aqueous solution are protected against “the action of oxygen,” is irrelevant because Petitioner does not explain how this protection would related to tableting compression forces. *Id.* at 38 (citing Ex. 1022, 19; Ex. 2013 ¶ 67). Patent Owner further argues that the statements in Szejtli regarding pressure and sensitivity are not comparable because they are in comparison to fragile microcapsules and may not correspond to the properties of 1-MCP. *Id.* at 38–39 (citing Ex. 1022, 20 (Table II); Ex. 2013 ¶¶ 67–68).

Patent Owner alleges that Petitioner’s reliance on Robinson (Ex. 1015) is misplaced because it does not relate to “tableting cyclodextrin complexes, tableting gaseous active ingredients, or tableting any compound relevant to 1-MCP.” *Id.* at 39 (citing Ex. 1015, 1:51–66, 6:29–56; Ex. 2013 ¶ 69).

Finally, Patent Owner argues that none of these references is a disclosure that would have created a reasonable expectation of success in the skilled artisan about releasing 1-MCP gas from a tablet. *Id.* at 40 (citing Ex. 2013 ¶ 70).

We are not persuaded. Sisler '988, Szejtli, and Robinson do not need to have disclosed use of or tableting of complexed 1-MCP to be relevant teachings for the person of ordinary skill in the art. Each reference would have been reviewed by the artisan for what it teaches or suggests about the field in general. *In re Mouttet*, 686 F.3d at 1331. In addition, as Petitioner argues, Szejtli's teachings relate to the properties of cyclodextrins as a class, and thus directly bear on 1-MCP. *See* Reply at 20–21.

Patent Owner has not persuaded us that the evidence of record does not support the skilled artisan's reasonable expectation of success in creating the claimed tablet dosage forms and belief that they would release 1-MCP gas when exposed to a solvent or water. For the reasons discussed herein, we find that claims 3 and 11 would have been obvious over Hisano and Daly.

F. Ground 3 – Obviousness over Daly and Hisano

Petitioner argues that Daly in view of Hisano teaches the subject matter of claims 3 and 11. Pet. 53–58. Petitioner incorporates its allegations regarding the teachings of Hisano and Daly above, with one exception: Petitioner argues that encapsulated 1-MCP gas

is delivered to plants by contacting the powder complex with a solvent, which liberates the methylcyclopropene from its encapsulation and allows it to contact the plant . . . the preferred solvent for use in this method of delivery is water. . . . more specifically, Daly teaches that a gas should be bubbled through the solvent (e.g., water) when it is in contact with the powder complex.

Id. at 53–54 (citing Ex. 1005, 7:60–67, 8:12–15). Petitioner argues that the skilled artisan using the teachings of Daly would “have been motivated to start with Daly’s teaching of 1-MCP and chosen to package the ingredient in an effervescent tablet based on the teachings of Hisano.” *Id.* at 54, 56–58. Petitioner argues that the skilled artisan would have had a reasonable expectation of success in combining the teachings to reach the claimed subject matter, incorporating the same arguments from Ground 2. *Id.* at 58. Petitioner’s expert supports this alternative obviousness position. Ex. 1006 ¶¶ 127–148.

Patent Owner does not dispute that the combination of Daly and Hisano teaches the claim limitations recited in claims 3 and 11. *See generally* PO Resp. We find that, based on the cited teachings as set forth above, Petitioner has sufficiently shown that each of the limitations of claims 3 and 11 is taught by the disclosures of Hisano and Daly.

With regard to the motivation of the skilled artisan to have made the recited tablet dosage and the artisan’s reasonable expectation of success in doing so, Patent Owner argues that Ground 3 fails for the same reasons as Ground 2. PO Resp. 41–44. Patent Owner argues that Petitioner’s reference to Daly’s teaching of “bubbling a gas through a solvent does not correct or in any way mitigate the deficiencies” Patent Owner addressed in Ground 2 regarding motivation to combine Hisano and Daly or a reasonable expectation of success in incorporating Daly’s 1-MCP encapsulation complex into a tablet formulation and releasing the agent from the tablet. *Id.* at 41–42 (citing Ex. 2013 ¶¶ 71–77). Dr. Beaudry opines that the skilled artisan “would not have been any more motivated to arrive at an effervescent tablet containing 1-MCP encapsulation complexes if she started with Daly’s disclosures” because the artisan, already facing the challenge of releasing

1-MCP from Daly's encapsulation complex would not have wanted to inhibit that release even further, which is what she would have expected tableting to do." Ex. 2013 ¶ 72. Dr. Beaudry argues that Dr. Walton has not identified any teaching in any cited reference suggesting that including an effervescent ingredient "would help drive a gaseous agent like 1-MCP out of an α -cyclodextrin complex." *Id.* ¶ 74.

With regard to the carbonic gas that Hisano discloses is formed during dissolution of the tablet form, Dr. Beaudry opines that the skilled artisan would have been "concerned about the effect of this release of an acid-forming gas when using Daly's encapsulated 1-MCP to treat plants" because the acid-forming gas would have lowered the pH and thus hindered release of the gas from the molecular encapsulation complexes in the tablet. *Id.* ¶ 75 (citing Ex. 2017, 6). Dr. Beaudry opines that the skilled artisan would not have understood Hisano's tablet "to be applicable for large-scale postharvest treatment of crops or flowers, where treating plants individually is inefficient and expensive" and thus would not have been motivated to combine the teachings to arrive at the claimed invention. *Id.* ¶ 77.

Finally, Patent Owner argues that statements in Petitioner's predecessor's patent application regarding alleged unexpected success of a similar invention show the lack of a reasonable expectation of success in this case. PO Resp. 43.

We are not persuaded. For the reasons discussed above, we find that the skilled artisan would have been motivated to start with the Daly's teachings of 1-MCP in cyclodextrin-caged, powdered form, and chosen to package the 1-MCP in an effervescent tablet as taught by Hisano for the improved dissolution benefits realized by use of an effervescent ingredient. In doing so, we further conclude that the artisan would have had a

reasonable expectation of success in making the subject matter of claims 3 and 11, which do not recite a specific amount of release of active agent or require application to large-scale crops. We are thus unpersuaded by Dr. Beaudry's concerns that the artisan would have found further potential inhibition – to the extent inhibition was observed – to be a deterrent to making the claimed subject matter. For the same reason, a potential drop in pH causing hindrance of gas release is unpersuasive.

CONCLUSION⁸

For the reasons discussed herein, we find that claims 3 and 11 would have been obvious over Daly and Hisano. In summary:

Claims	35 U.S.C. §	References/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
3, 11	103(a)	Hisano, Daly	3, 11	
3, 11	103(a)	Daly, Hisano	3, 11	
Overall Outcome			3, 11	

ORDER

For the reasons given, it is:

⁸ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

ORDERED that Petitioner has shown based on a preponderance of evidence that claims 3 and 11 of U.S. Patent No. 6,897,185 B1 are unpatentable; and

FURTHER ORDERED that, because this is a final written decision, the parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2021-00451
Patent 6,897,185 B1

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